CRISP: Cross-Modal Residual Guidance and Spatial Realignment for Remote Sensing Visual Question Answering

Changhui Xu, Zhongle Ren, *Member, IEEE*, Biao Hou, *Senior Member, IEEE*, Cheng Zhang, Jiawei Ning, Weibin Li, *Senior Member, IEEE*, and Licheng Jiao, *Life Fellow, IEEE*

***Abstract*—Remote sensing visual question answering (RSVQA) remains challenging because the evidence relevant to the question in remote sensing (RS) imagery is typically sparse, spatially dispersed, and highly variable in scale and layout. This structural heterogeneity poses a major challenge to existing transfer strategies, which often struggle to achieve effective query-conditioned semantic focusing and localized spatial refinement. To address this issue, we propose CRISP, a task-specific parameter-efficient adaptation framework for RSVQA built on a frozen ViLT backbone. CRISP comprises two complementary components. First, a cross-modal residual guidance (CMRG) module generates instance-specific guidance tokens from pooled image and question summaries, steering early cross-modal interaction toward query-relevant content while suppressing background interference. Second, an attention-guided spatial realignment (ASR) module performs offset-guided feature realignment within intermediate Transformer layers, enabling localized refinement of spatial evidence under scale variation and sparse semantic distribution. Extensive experiments on the RSVQA-LR and RSVQA-HR benchmarks show that CRISP achieves strong overall performance and consistently improves overall accuracy (OA) and average accuracy (AA) over prior methods, with particularly notable gains on presence, comparison, and region-related questions. These results demonstrate that residual guidance and spatial realignment together provide an effective task-specific parameter-efficient adaptation strategy for RSVQA under the frozen-backbone setting. The code will be available at https://github.com/PhD-Xu/CRISP**

***Index Terms*—Cross-modal guidance, remote sensing (RS), spatial realignment, visual question answering (VQA).**

I. Introduction

DRIVEN by recent advancements in deep learning and the explosive growth of remote sensing (RS) data, RS technology has emerged as an indispensable tool for diverse real-world applications, such as ocean environment monitoring, land resource and ecosystem assessment, cryosphere change observation, and disaster or anomaly detection [1]. Early research on RS image (RSI) interpretation primarily concentrated on single-modal visual tasks, including object detection [2], [3], [4], image fusion [5], [6], semantic segmentation [7], [8], [9], scene classification [10], [11], [12], and change detection [13], [14], [15]. These approaches rely exclusively on visual cues to characterize object categories and spatial distribution. These purely visual approaches often struggle to model the high-level semantic relationships and contextual reasoning required for comprehensive interpretation of RSI. Recently, with the rapid advancement of large-scale vision-language pretraining and large language models (LLMs) [16], RSI understanding has evolved toward multimodal vision-language tasks, including image captioning [17], [18], [19], visual question answering (VQA) [20], [21], [22], visual grounding [23], [24], [25], and image-text retrieval [26], [27], [28], offering a new paradigm for semantic-level understanding in RS.

In the domain of natural images, multimodal vision-language tasks have achieved significant progress driven by large-scale pretrained models such as CLIP [29], BLIP [30], and ViLT [31]. In this context, VQA stands out as a representative task that jointly processes images and natural-language queries to produce question-conditioned answers, thereby facilitating fine-grained cross-modal semantic reasoning and knowledge integration. Building on these advances, researchers have extended the VQA paradigm to RS scenarios, giving rise to remote sensing visual question answering (RSVQA). RSVQA targets refined scene understanding and interactive semantic reasoning over complex RSI. However, directly transferring VQA models from natural images to RS domains remains a very challenging task [32]. RSI is characterized by significant scale variation, sparse semantic distribution, and complex spatial layouts [33]. These complexities render standard visual representation learning inefficient and make cross-modal alignment substantially more difficult, particularly when reasoning over fine-grained objects that are sparsely dispersed across large scenes. Most existing RSVQA approaches rely on fully fine-tuning large-scale pretrained models to achieve task adaptation. This paradigm incurs high computational costs and is often infeasible under limited

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The authors are with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China, Joint International Research Laboratory of Intelligent Perception and Computation, Xidian University, Xi'an 710071, China (e-mail: xch0113@stu.xidian.edu.cn; zlren@xidian.edu.cn; avcodec@hotmail.com; zcsisiyao@163.com; jwning@stu.xidian.edu.cn; weibinli@xidian.edu.cn; lchjiao@mail.xidian.edu.cn).

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RS annotations [34]. Although lightweight parameter-efficient fine-tuning (PEFT) strategies such as prompt-based tuning and adapter-based tuning [35], [36], [37], [38] have shown effectiveness in natural-image tasks, they are less suited to RSVQA [22], [39]. RS scenes exhibit strong scale variation and sparse, spatially dispersed evidence, while many existing PEFT designs rely on sample-agnostic conditioning signals and spatially invariant feature adaptation. This structural mismatch restricts the model's flexibility to accommodate the pronounced scale variation, sparse semantic distribution, and complex spatial layouts of RSI, thereby limiting fine-grained cross-modal reasoning in heterogeneous RS scenes.

This mismatch becomes more evident in RSVQA at two levels. Existing lightweight designs often fail to provide effective query-conditioned semantic focusing at the input stage and lack localized spatial refinement within intermediate representations. In RSVQA, question-relevant evidence is often sparse, locally distributed, and highly variable in scale and position across large RS scenes. As a result, effective reasoning requires not only early semantic focusing conditioned on the query, but also localized spatial realignment during feature propagation. Such requirements are difficult to satisfy with static conditioning and spatially uniform adaptation alone. Consequently, existing lightweight designs often struggle to suppress irrelevant background regions at the input stage and to refine question-relevant spatial evidence within intermediate layers.

To address these limitations, we propose CRISP, task-specific parameter-efficient adaptation framework for RSVQA built upon a frozen ViLT backbone. Specifically, the framework incorporates a cross-modal residual guidance (CMRG) module, which synthesizes instance-specific guidance tokens from pooled image and question summaries to steer early cross-modal interaction toward query-relevant content. In parallel, CRISP introduces an attention-guided spatial realignment (ASR) module, which performs offset-guided feature realignment within Transformer layers to better accommodate scale variation, sparse semantic distribution, and complex spatial layouts in RSI. By updating only the guidance generator, spatial realignment modules, and answer prediction head, the proposed framework improves semantic alignment and spatial feature refinement while maintaining parameter efficiency. Technically, CMRG generates residual guidance tokens conditioned on global visual and textual summaries, whereas ASR equips each ViLT layer with an attention-guided offset generator (AGOG) and a dynamic resampler (DRS) for localized feature realignment. The combination of CMRG and ASR provides complementary benefits for question-conditioned reasoning in heterogeneous RS scenes.

The main contributions of this article can be summarized as follows.

1) We propose CRISP, a task-specific parameter-efficient adaptation framework for RSVQA that integrates CMRG with ASR on top of a frozen ViLT backbone. The novelty mainly lies in the lightweight design and coordinated integration of task-specific modules for frozen-backbone RSVQA adaptation, which helps alleviate the structural mismatch between insufficient query-conditioned semantic focusing at the input stage and inadequate localized spatial refinement in intermediate layers.
2) We develop a CMRG module to generate instance-specific guidance tokens from pooled image and question summaries. This design provides query-conditioned semantic guidance for early cross-modal interaction and helps suppress background interference in large RS scenes.
3) We introduce an ASR module to perform offset-guided feature realignment within Transformer layers. This module enhances localized feature refinement under scale variation and sparse semantic distribution, and works complementarily with CMRG to support fine-grained cross-modal reasoning in heterogeneous RS scenes.

The rest of this article is organized as follows. Section II reviews the recent advances in VQA for both natural images and RSI. Section III presents the proposed CRISP framework in detail, including the CMRG and ASR modules. Section IV reports the experimental results and comparative analyses. Finally, Section V summarizes the conclusions and discusses potential directions for future research.

II. Related Work

This section briefly reviews related work on VQA in both natural-image and RSI, with a particular focus on lightweight transfer and adaptation strategies.

A. Visual Question Answering

VQA stands as one of the core tasks in the field of multimodal artificial intelligence, aiming to enable machines to generate accurate natural language answers based on input images and natural language questions. Antol et al. [40] pioneered the development of benchmark datasets and system model frameworks for VQA, spurring extensive subsequent research aimed at addressing its multifaceted challenges. Notably, most follow-up studies in VQA have focused on designing innovative architectures, particularly emphasizing the modeling of semantic associations between visual and textual patterns. Advances in these architectures have provided crucial technical foundations for tackling core bottlenecks in VQA tasks. For instance, Chen et al. [41] effectively enhanced information fusion capabilities through a rational knowledge-aware attention fusion decoder. Peng and Li [42] strengthened the semantic relevance of visual features via an intramodal and intermodal interaction module. Meanwhile, Zhang and Zhang [43] addressed the issue of weak visual-text alignment by achieving fine-grained alignment through prompt-guided visual encoders and optimizing coarse-grained alignment with a Model-of-Events connector.

With the advent of large-scale vision-language pretraining, VQA has benefited from stronger cross-modal priors and more unified representations, improving robustness while reducing reliance on task-specific feature engineering [29], [30], [31]. However, transferring such pretrained backbones to new datasets or application scenarios often incurs substantial computational cost when full fine-tuning is adopted, and the learned cross-modal alignment can be sensitive to

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distribution shifts. As a result, lightweight adaptation strategies have attracted increasing attention, where only a small number of additional parameters are optimized while the backbone remains frozen [44]. This paradigm provides a practical foundation for domain-specific extensions of VQA by reducing the optimization burden and mitigating overfitting under limited supervision. Nevertheless, many existing lightweight designs rely on static conditioning signals and spatially uniform feature transformation, which limits their ability to adapt to content-dependent visual semantics and spatial variability. This motivates more instance-aware guidance and localized feature adaptation mechanisms that can better support question-conditioned reasoning while keeping most backbone parameters frozen.

B. Remote Sensing Visual Question Answering

With the rapid advancement and technological maturity of VQA tasks in natural image domains, researchers in RS have also begun to turn their attention to this cross-domain research direction and progressively explore the adaptive validation and application expansion of VQA technology within RSI scenarios. Addressing the inherent resolution heterogeneity of RSI, Lobry et al. [45] pioneered the transfer of VQA techniques to RS by constructing a comprehensive RSVQA dataset that encompasses both low-resolution and high-resolution samples, containing nearly 15 million samples, and by establishing the first systematic benchmark framework for this emerging field. Subsequently, this dataset and framework became the cornerstone of subsequent research, on which numerous researchers iteratively proposed a series of targeted optimization strategies. In the technical exploration of the RSVQA task, early studies mainly focused on optimizing feature extraction and cross-modal fusion. Yuan et al. [46] designed the cross-modal global attention module and cross-modal spatial transformer module to enable efficient extraction of global semantic and multiscale regional features and to adapt to the significant scale heterogeneity of RS targets. Zhang et al. [21] introduced hash positional embedding and a multiscale feature extraction mechanism to enhance cross-modal feature alignment accuracy, and constructed a question-guided relation network unit to model associative relationships between objects in RS scenes. Sarkar et al. [47] integrated the segment anything model into the RSVQA workflow, providing precise spatial constraints for regional feature extraction by automatically segmenting target areas in RSI. Feng et al. [48] proposed a hierarchical feature extraction and cross-scale fusion strategy with a bidirectional fusion mechanism to coordinate multiscale information and enhance feature representation completeness.

As RSVQA progresses beyond early designs centered on feature extraction and cross-modal fusion, recent efforts have increasingly examined how generic vision-language pretrained backbones can be transferred to RS. However, the domain gap caused by different imaging mechanisms, target distribution, and background complexity often leads to suboptimal cross-modal alignment and limited generalization on RSVQA benchmarks. To alleviate this bottleneck under constrained training budgets, representative studies have explored lightweight cross-domain adaptation and domain knowledge augmentation. Wang and Ghamisi [22] proposed RSAdapter by inserting lightweight adaptation layers into pretrained models to reduce fine-tuning cost while improving RS adaptability. Zhu et al. [49] introduced a geography knowledge graph-based framework to inject domain-specific knowledge and improve geographic reasoning accuracy. Zhang et al. [50] proposed a frequency-domain transfer strategy with a primary frequency adaptation module that learns frequency-specific representations to better match the spectral characteristics of RS data, thereby reducing the generalization drop when natural-image pretrained models are directly applied to RSVQA.

More recently, RS large vision-language models have attracted increasing attention. GeoChat [51] introduces a grounded large vision-language model for RS, enabling image-level dialog, region-level interaction, and visual grounding. VHM [52] develops a versatile and honest vision-language model for RSI analysis by leveraging large-scale RSI-text data and instruction data. SkySenseGPT [53] further constructs a fine-grained instruction-tuning dataset and model to enhance RS vision-language understanding. These studies demonstrate the potential of large-scale RS vision-language modeling for general multimodal interpretation. Different from these large-scale generative or instruction-following frameworks, our CRISP focuses on parameter-efficient adaptation for RSVQA by introducing lightweight CMRG and spatial realignment modules on top of a frozen vision-language backbone.

Collectively, these studies highlight the value of lightweight transfer, domain-aware cues, and large-scale RS vision-language modeling, while also indicating that RSVQA still requires more effective query-conditioned guidance at the input stage and more localized spatial refinement within intermediate representations.

III. Methodology

Sections III-A–III-E present the overall architecture, the pretrained model, the CMRG module, the ASR module, and the answer-prediction procedure.

A. Overall Architecture

The overall architecture of the proposed CRISP framework is illustrated in Fig. 1. Built upon a pretrained ViLT backbone, CRISP takes an RSI and its question as input, projects the image into patch embeddings and the question into token embeddings, and encodes the fused multimodal sequence while keeping the backbone frozen. Two lightweight trainable modules are introduced to enable parameter-efficient and semantically adaptive RSVQA. Specifically, the CMRG module generates content-aware guidance tokens conditioned on the image and question context, and prepends them to the textual stream to provide query-conditioned guidance for early cross-modal interaction. In parallel, the ASR module performs offset-guided feature realignment within Transformer layers to refine spatial representations and better accommodate RS-specific scene variability. The resulting multimodal representation is finally fed into a lightweight classifier for answer prediction, while only the CMRG module, the ASR module, and the prediction head are updated during training to preserve efficiency.

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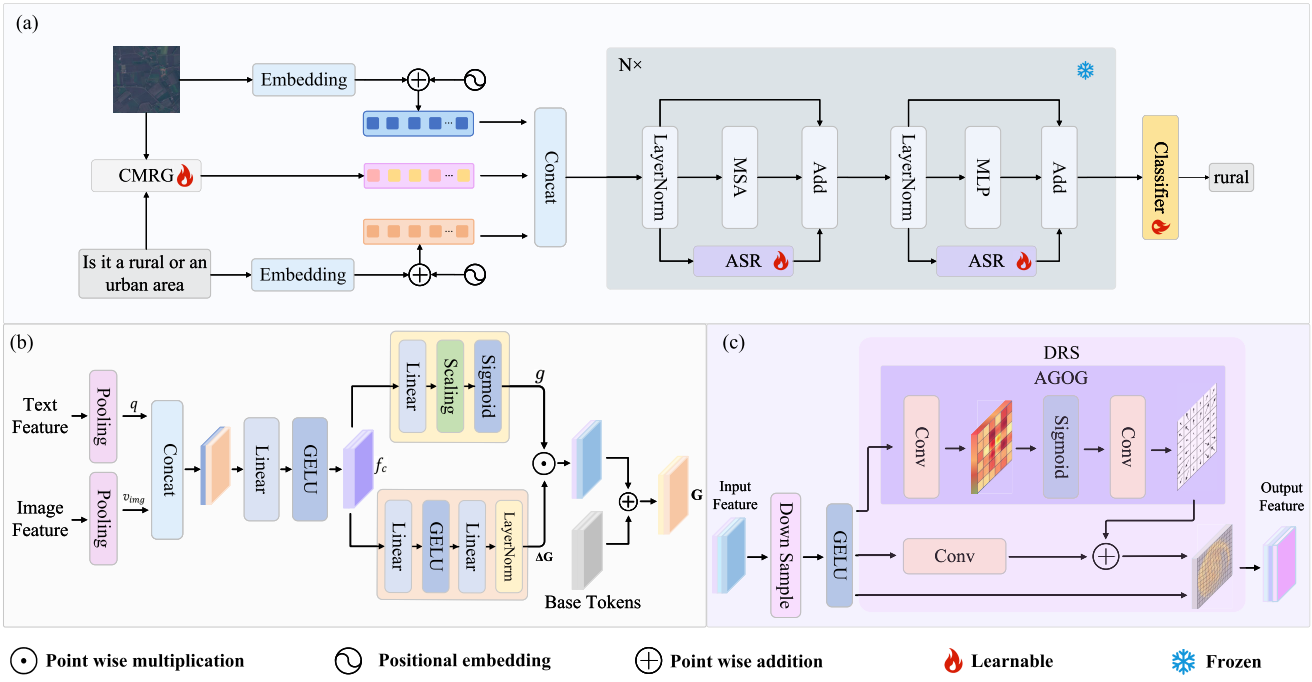


Fig. 1. Overall architecture of the proposed CRISP framework. (a) Overview of the CRISP network. (b) Structure of the CMRG module. (c) Details of the ASR module. Components labeled as "frozen" are kept fixed during fine-tuning, whereas those labeled as "learnable" are updated.

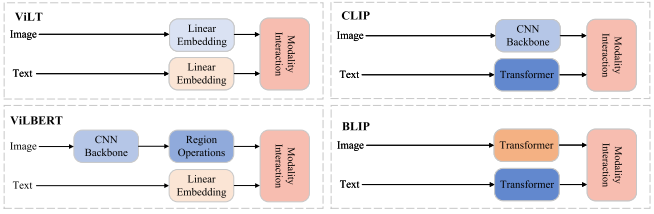


Fig. 2. Comparison of representative vision-language backbones. ViLT uses linear patch projection and single-stream early fusion, whereas ViLBERT, CLIP, and BLIP rely on CNN/region features or dual encoders. ViLT therefore offers a more compact architecture with lower training and inference overhead.

B. Pretrained Model

ViLT is selected as the backbone to balance multimodal reasoning capacity with practical computational constraints. As summarized in Fig. 2, representative vision-language backbones differ primarily in visual encoding and fusion strategy. ViLBERT depends on a convolutional visual backbone and region-level operations before modality interaction, whereas CLIP and BLIP typically follow dual-encoder pipelines that encode image and text separately and then rely on additional interaction modules for downstream reasoning. In contrast, ViLT adopts a single-stream early-fusion Transformer that projects image patches into token embeddings and performs unified multimodal modeling without region features or extra convolutional stems, resulting in a more compact architecture with reduced parameters and lower training and inference overhead, which is well aligned with RS settings where annotations and compute budgets are often limited. Building upon the frozen backbone, the proposed CRISP framework is constructed by introducing two lightweight trainable components, namely the CMRG module and the ASR module, to enable query-conditioned cross-modal guidance and localized spatial refinement while retaining the efficiency advantages of the backbone.

Let question token embeddings be $\mathbf{T} \in \mathbb{R}^{(N_t+1)\times d}$ and visual patch embeddings be $\mathbf{V} \in \mathbb{R}^{(N_v+1)\times d}$, where N_t and N_v denote the number of textual tokens and image patches, respectively, and d is the hidden dimension. The learnable positional embeddings for text and vision are denoted as $\mathbf{T}_{\text{pos}}$ and $\mathbf{V}_{\text{pos}}$, and the modality type embeddings are denoted as $\mathbf{t}_{\text{type}}$ and $\mathbf{v}_{\text{type}}$. The fused input sequence is $\mathbf{X}_0 \in \mathbb{R}^{(N_t+N_v+2)\times d}$. Specifically

$$\mathbf{T} = \left[\mathbf{t}_{\text{cls}}; \mathbf{t}_1; \ldots; \mathbf{t}_{N_t}\right] + \mathbf{T}_{\text{pos}} \tag{1}$$

$$\mathbf{V} = \left[\mathbf{v}_{\text{cls}}; \mathbf{v}_1; \ldots; \mathbf{v}_{N_v}\right] + \mathbf{V}_{\text{pos}} \tag{2}$$

$$\mathbf{X}_0 = \left[\mathbf{T} + \mathbf{t}_{\text{type}}; \mathbf{V} + \mathbf{v}_{\text{type}}\right] \tag{3}$$

where $\mathbf{t}_i$ and $\mathbf{v}_j$ are the content embeddings of the ith token and the jth patch, and $[\cdot;\cdot]$ denotes concatenation along the token dimension.

Given $\mathbf{X}_0$, the backbone encodes it with N transformer layers

$$\mathbf{X}'_\ell = \mathbf{X}_{\ell-1} + \text{MSA}\left(\text{LN}\left(\mathbf{X}_{\ell-1}\right)\right)$$
$$\mathbf{X}_\ell = \mathbf{X}'_\ell + \text{MLP}\left(\text{LN}\left(\mathbf{X}'_\ell\right)\right), \quad \ell = 1, \ldots, N \tag{4}$$

where LN$(\cdot)$ denotes layer normalization, MSA$(\cdot)$ denotes multihead self-attention, and MLP$(\cdot)$ denotes the position wise feed forward network. $\mathbf{X}'_\ell$ denotes the intermediate representation after the self-attention sublayer.

C. Cross-Modal Residual Guidance

RSVQA often requires reasoning over a small set of question-dependent cues that occupy limited spatial regions and are sparsely distributed across large-scale scenes. Existing lightweight conditioning strategies commonly adopt static

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textual tokens shared across all samples, making the guidance insensitive to both scene content and question intent. This limitation can diffuse cross-modal attention and increase the attention assigned to irrelevant background regions, which undermines fine-grained semantic alignment and subsequent reasoning. As illustrated in Fig. 1, the CMRG module mitigates this issue by generating content-aware guidance tokens conditioned on a pooled global image descriptor and a pooled question representation. The synthesized guidance tokens are prepended to the textual sequence, enabling instance-specific semantic guidance that strengthens cross-modal interaction and improves question-aligned representation learning.

1) Question and Image Summaries: Given the question token embeddings $\mathbf{T}$ and the frozen visual patch embeddings $\mathbf{V}$ from the pretrained model subsection, the question intent and a global visual summary are first compressed into two vectors. A lightweight attention-based pooling operator Pool$(\cdot)$ is applied along the token dimension. The same pooling operator, parameterized by a shared learnable vector $\mathbf{w}_p$, is applied to both the question and image token sequences

$$\mathbf{q} = \text{Pool}(\mathbf{T}) \in \mathbb{R}^d \tag{5}$$

$$\mathbf{v}_{\text{img}} = \text{Pool}(\mathbf{V}) \in \mathbb{R}^d. \tag{6}$$

For clarity, Pool$(\cdot)$ can be implemented as a weighted sum

$$\text{Pool}(\mathbf{Z}) = \sum_i \alpha_i \mathbf{z}_i, \qquad \alpha_i = \frac{\exp\left(\mathbf{w}_p^\top \mathbf{z}_i\right)}{\sum_j \exp\left(\mathbf{w}_p^\top \mathbf{z}_j\right)} \tag{7}$$

where $\mathbf{Z} = \{\mathbf{z}_i\}$ denotes a token sequence and $\mathbf{w}_p$ is a learnable vector shared by the text and image modalities. This step produces compact summaries that preserve global semantics with negligible additional cost. The resulting $\mathbf{q}$ and $\mathbf{v}_{\text{img}}$ are then used to generate the CMRG tokens.

2) Fusion Context: The pooled question summary $\mathbf{q}$ and the global image descriptor $\mathbf{v}_{\text{img}}$ are fused into a semantic context vector $\mathbf{f}_c \in \mathbb{R}^d$ through a lightweight projection with GELU activation

$$\mathbf{f}_c = \text{GELU}\left(\mathbf{W}_f \left[\mathbf{v}_{\text{img}}; \mathbf{q}\right] + \mathbf{b}_f\right) \tag{8}$$

where $[\cdot;\cdot]$ denotes vector concatenation and $\mathbf{W}_f$ and $\mathbf{b}_f$ are learnable parameters. The resulting $\mathbf{f}_c$ serves as a joint conditioning signal that couples visual context with question intent.

3) Residual Guidance Synthesis: Conditioned on $\mathbf{f}_c$, a residual guidance tensor $\Delta\mathbf{G} \in \mathbb{R}^{K\times d}$ and a token-wise gate $\mathbf{g} \in (0,1)^K$ are predicted for K guidance tokens

$$\Delta\mathbf{G} = \text{Reshape}\left(\mathbf{W}_2\, \text{GELU}\left(\mathbf{W}_1 \mathbf{f}_c + \mathbf{b}_1\right) + \mathbf{b}_2\right) \tag{9}$$

$$\mathbf{g} = \text{Sigmoid}\left(\frac{1}{\tau}\mathbf{W}_g \mathbf{f}_c + \mathbf{b}_g\right) \tag{10}$$

where τ is a temperature scalar. Let $\mathbf{G}_0 \in \mathbb{R}^{K\times d}$ denote learnable base guidance tokens. The final guidance tokens are obtained by gated residual composition

$$\mathbf{G} = \mathbf{G}_0 + \left(\mathbf{g}\mathbf{1}_d^\top\right) \odot \Delta\mathbf{G} \tag{11}$$

where τ is a learnable temperature scalar initialized to 1.0 and jointly optimized with the other trainable parameters in the CMRG module. It controls the smoothness of the gating activation and allows the model to adaptively adjust the sharpness of the token-wise residual guidance.

4) Content-Aware Guidance Augmentation: The synthesized guidance tokens $\mathbf{G}$ are prepended to the textual sequence to form an extended text token sequence $\mathbf{T}_e$. In our implementation, the guidance tokens are treated as additional textual guidance tokens. Therefore, they are assigned dedicated positional embeddings and the text modality type embedding before being prepended to the original textual tokens. The extended text tokens are then concatenated with visual tokens to construct the guidance-augmented multimodal input $\mathbf{X}_0^e$, which is fed into the subsequent Transformer encoder

$$\mathbf{T}_e = \left[\mathbf{G} + \mathbf{G}_{\text{pos}} + \mathbf{t}_{\text{type}}; \mathbf{T}\right], \qquad \mathbf{X}_0^e = [\mathbf{T}_e; \mathbf{V}] \tag{12}$$

where $[\cdot;\cdot]$ denotes concatenation along the token dimension, $\mathbf{G}_{\text{pos}} \in \mathbb{R}^{K\times d}$ denotes the dedicated positional embeddings assigned to the K guidance tokens, and $\mathbf{t}_{\text{type}} \in \mathbb{R}^d$ denotes the text modality type embedding used for the guidance tokens. Overall, the CMRG module generates content-aware guidance tokens that provide instance-specific semantic guidance for question-aligned cross-modal interaction while keeping the number of additional trainable parameters lightweight.

D. Attention-Guided Spatial Realignment

As shown in Fig. 1, the ASR module is introduced to better accommodate the characteristic RS challenges in RSVQA, where question-relevant evidence is often sparse and spatially dispersed, and targets can exhibit pronounced scale variation across large scenes. ASR augments each Transformer block with a lightweight parallel branch that performs attention-guided, spatially variant feature realignment. Given an intermediate feature map $\mathbf{X} \in \mathbb{R}^{B\times C\times H_f\times W_f}$, it predicts a group-wise 2-D offset field from attention responses, resamples $\mathbf{X}$ through a differentiable sampler to emphasize informative neighborhoods, and injects the aligned features back into the residual stream with a learnable scaling factor. The module is inserted in parallel to both the self-attention and feed-forward sublayers while keeping the backbone frozen, enabling content-adaptive localized spatial refinement with only a minor increase in trainable parameters.

1) Offset Prediction: Let AGOG$(\cdot)$ denote the AGOG, which predicts spatial offset modulation conditioned on contextual features through a lightweight attention mechanism. Given an intermediate feature map $X \in \mathbb{R}^{B\times C\times H_f\times W_f}$, we first apply a preliminary projection operator $\mathcal{R}(\cdot)$ to obtain the feature representation used for offset prediction

$$Z = \mathcal{R}(X) = \phi_r(X) \tag{13}$$

where $\phi_r(\cdot)$ is a 1×1 convolutional projection with channel-reduction ratio r. The spatial resolution is kept unchanged, and thus $Z \in \mathbb{R}^{B\times C/r\times H_f\times W_f}$. Based on Z, AGOG first produces a group-wise spatial attention map and then maps it to an attention-guided offset modulation field

$$\Delta O = \text{AGOG}(Z) \tag{14}$$

where $\Delta O \in \mathbb{R}^{B\times 2G\times H_f\times W_f}$ denotes a group-wise 2-D offset modulation field, and G is the number of channel groups. The final offset field used for dynamic resampling is obtained by combining the base offset field and the attention-guided modulation

$$O = O_{\text{base}} + \Delta O \tag{15}$$

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where O_{base} denotes the base offset field produced by the dynamic resampling branch. Each group contains two offset channels corresponding to the horizontal and vertical directions.

2) Dynamic Resampling: In RSI, question-relevant evidence is often spatially sparse, and targets can appear at markedly different scales across large scenes. Therefore, a content-adaptive and spatially variant resampling strategy is adopted, in which per-location offsets guide a differentiable sampler. The DRS is defined as

$$\text{DRS}(\mathbf{X}, \mathbf{O}) = \text{GridSample}\left(\text{Group}(\mathbf{X}), \mathcal{G}(\mathbf{O})\right) \tag{16}$$

followed by a shrink and expand pathway

$$\mathbf{Y}_{\text{down}} = \text{Linear}(\mathbf{X}) \tag{17}$$

$$\mathbf{Y}_{\text{up}} = \text{DRS}\left(\text{GELU}\left(\psi_{\text{down}}(\mathbf{Y}_{\text{down}})\right), \mathbf{O}_{\text{up}}\right) \tag{18}$$

where $\mathbf{O}, \mathbf{O}_{\text{up}} \in \mathbb{R}^{B\times 2G\times H_f\times W_f}$ denote group-wise offset fields in pixel units, and each group has two channels corresponding to horizontal and vertical offsets. The operator Group$(\cdot)$ splits channels into G groups so that each group uses its own sampling grid. The normalized grid $\mathcal{G}(\mathbf{O})$ is constructed from base centers $\mathcal{G}_0$ and normalized offsets as

$$\mathcal{G}(\mathbf{O}) = \mathcal{G}_0 + \text{norm}(\mathbf{O})$$

$$\text{norm}(\mathbf{O}) = \left(\frac{2}{W_f - 1}O_x, \frac{2}{H_f - 1}O_y\right) \tag{19}$$

in which (O_x, O_y) denote the horizontal and vertical components of $\mathbf{O}$. The operator GridSample$(\cdot)$ is differentiable, Linear$(\cdot)$ is a learnable projection, and ψ_{down} is a 1×1 projection. Overall, the attention-guided offsets and spatially variant resampling provide localized feature realignment, which improves spatial feature refinement under scale variation and sparse evidence in RSVQA while keeping the additional trainable parameters lightweight.

E. Answer Predicting

Let $\mathbf{X}_L$ denote the output sequence of the last Transformer encoder layer in the ViLT backbone, where d is the hidden dimension. The global multimodal embedding is constructed as

$$\mathbf{f} = \left[\mathbf{x}_{\text{cls}}; \mathbf{q}; \mathbf{v}_{\text{img}}\right] \in \mathbb{R}^{3d} \tag{20}$$

where $\mathbf{x}_{\text{cls}} \in \mathbb{R}^d$ denotes the final hidden state of the `[CLS]` token, and $\mathbf{q} \in \mathbb{R}^d$ and $\mathbf{v}_{\text{img}} \in \mathbb{R}^d$ denote the pooled question and image summaries obtained by applying the pooling operator in (5)–(7) to the question and visual embeddings, respectively. For fair comparison, $\mathbf{q}$ and $\mathbf{v}_{\text{img}}$ are computed in all settings, including the baseline and w/o CMRG variants. The difference is that only the full model uses $(\mathbf{q}, \mathbf{v}_{\text{img}})$ to generate residual guidance tokens through CMRG, whereas the ablation variants use these summaries only as auxiliary features for answer prediction without guidance generation. A three-layer MLP classifier is applied to $\mathbf{f}$ to produce logits

$$\mathbf{h}_1 = \text{Dropout}(\text{GELU}(\mathbf{W}_1\mathbf{f} + \mathbf{b}_1)) \tag{21}$$

$$\mathbf{h}_2 = \text{Dropout}(\text{GELU}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2)) \tag{22}$$

$$\mathbf{z} = \mathbf{W}_3\mathbf{h}_2 + \mathbf{b}_3 \tag{23}$$

where $\mathbf{W}_1$, $\mathbf{W}_2$, $\mathbf{W}_3$ and $\mathbf{b}_1$, $\mathbf{b}_2$, $\mathbf{b}_3$ are learnable parameters, and $\mathbf{z} \in \mathbb{R}^{N_{\text{ans}}}$ with N_{ans} denoting the number of candidate answers. The predicted distribution over candidate answers is obtained by softmax normalization

$$\boldsymbol{\pi} = \text{Softmax}(\mathbf{z}) \tag{24}$$

and the final prediction is $\hat{y} = \arg\max_k \pi_k$, where π_k denotes the kth element of $\boldsymbol{\pi}$. The index $\hat{y}$ corresponds to the most probable answer class in the predefined answer vocabulary. During inference, $\hat{y}$ is mapped to its associated answer token as the final output.

IV. Experimental Study

A. Datasets

Experiments are conducted on two standard RSVQA benchmarks, RSVQA-LR and RSVQA-HR [45]. Both datasets are constructed from optical imagery paired with OpenStreetMap (OSM) annotations, and they provide complementary evaluation settings in terms of spatial resolution, scene complexity, and question types.

1) *RSVQA-LR*: This dataset is built from Sentinel-2 multispectral tiles acquired over the Nethers at a spatial resolution of 10 m ground sampling distance. From these tiles, 772 RGB image patches of size 256×256 pixels are extracted. Each image is associated with multiple automatically generated questions, resulting in 77 232 image question answer triplets. The questions are grouped into four categories, namely Count, Presence, Comparison, and Rural or Urban, covering object counting, existence queries, relational comparisons, and coarse scene type classification. We use the official predefined splits, with 77.8% for training, 11.1% for validation, and 11.1% for testing to evaluate model performance.
2) *RSVQA-HR*: This dataset contains 10 659 images of size 512×512 pixels at a spatial resolution of 15 cm. Each image covers approximately 5898 m^2. All images are extracted from the United States Geological Survey High Resolution Orthoimagery (HRO) dataset. The dataset comprises 1 066 316 image question answer triplets, with approximately 100 questions and corresponding answers per image. The questions are categorized into four types, including counting, presence, comparison, and area. The official split allocates 61.5% for training, 11.2% for validation, and 27.3% for testing. Notably, the test set includes two subsets, HR1 and HR2. HR1 has coverage similar to the training and validation sets, while HR2 evaluates generalization by using geographically disjoint coverage. In addition, HR2 uses data collected from different sensors.

Together, RSVQA-LR and RSVQA-HR span a wide range of spatial resolutions and semantic difficulties, providing a comprehensive benchmark to assess the parameter efficiency and cross-split evaluation performance of the proposed CRISP framework.

B. Experimental Settings

The proposed method is implemented in PyTorch based on the publicly available ViLT framework. The backbone is a

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ViLT encoder pretrained on large-scale image-text corpora and generic VQA datasets. In all experiments, the ViLT backbone is kept frozen and only the parameters of the proposed CMRG and ASR modules, together with the answer prediction head, are updated during training, ensuring parameter-efficient adaptation to the RSVQA benchmarks. All experiments are conducted on a single NVIDIA RTX 4090 GPU.

For each image-question pair, the RSI is resized to the ViLT input resolution and normalized using the standard mean and variance adopted in pretraining. When training on RSVQA-HR, the high-resolution aerial images are resampled to the same input resolution while preserving the aspect ratio. The question text is lowercased and tokenized using the WordPiece tokenizer of ViLT, and then truncated or padded to a fixed maximum length. Following common practice in RSVQA, answer prediction is formulated as a multiclass classification problem over an answer vocabulary constructed from the training split, and the same vocabulary is employed for validation and test.

Cross-entropy loss is employed, and the AdamW optimizer is used to update the trainable parameters. For RSVQA-LR, the initial learning rate is set to 5×10^{-4} with a batch size of 128. For RSVQA-HR, the initial learning rate is set to 1×10^{-4} with a batch size of 50. A cosine annealing schedule is adopted to decay the learning rate during training. All models are trained for 25 epochs on both datasets. Unless otherwise specified, identical optimization settings are used across all variants to ensure fair comparison. During training, overall accuracy (OA) on the validation set is monitored, and the checkpoint with the best validation performance is selected for final evaluation on the test set. Performance is reported using OA and average accuracy (AA). OA measures the percentage of correctly predicted answers over all test samples. AA is computed by averaging the accuracies of different question types, which reduces the influence of category imbalance and reflects per-type performance more fairly. All ablation variants share the same frozen ViLT backbone, tokenizer, answer vocabulary, optimizer, and training schedule, so that the observed differences can be primarily attributed to the proposed modules rather than to changes in the backbone or training setup. Unless otherwise specified, the number of guidance tokens is set to $K = 10$ in all main experiments. For the ASR module, we set the number of channel groups to $G = 2$ and the channel-reduction ratio in AGOG to $r = 8$. The channel-compression and attention-generation layers are initialized with Kaiming normal initialization, while the final offset-modulation layer is zero-initialized. The base offset convolution in the dynamic resampling branch is initialized from a normal distribution with standard deviation 0.001, and the residual scaling factors are initialized to 1.

C. Comparison Results and Analysis

In this section, we make a comparison of the proposed method against the current eight RSVQA approaches, listed as follows. Comparative studies are employed on RSVQA-LR and RSVQA-HR datasets.

1) *RSVQA [45]:* A baseline with a CNN image encoder and an RNN question encoder. The two features are fused and classified to predict answers.

TABLE I

COMPARISON WITH EXISTING METHODS ON THE RSVQA-LR DATASET. ALL VALUES ARE REPORTED AS PERCENTAGES (%). * DENOTES OPEN-SOURCE CODE TESTED INDEPENDENTLY

Method	Count	Presence	Comparison	Rural/Urban	AA	OA
RSVQA[45]*	66.02±0.92	89.34±0.10	83.28±1.56	92.67±1.15	82.83±0.80	80.08±0.88
EasyToHard[46]*	69.22±0.33	90.66±0.24	87.49±0.10	91.67±1.57	84.76±0.35	83.09±0.15
SHRNet[21]	73.87±0.22	91.03±0.13	90.48±0.05	94.00±0.87	87.34±0.13	85.85±0.28
UCAGAN[54]	70.68±0.48	91.17±0.23	89.08±0.20	91.00±1.00	85.48±0.58	83.95±0.22
SAMFPN[55]	69.01±0.20	91.21±0.31	91.01±0.21	92.67±1.53	86.10±0.41	84.76±0.13
Bi-Modal[56]	72.22±0.57	91.06±0.17	91.16±0.09	92.66±1.52	86.78±0.28	85.56±0.16
X-PFA[50]	**74.30±0.12**	90.80±0.27	91.71±0.04	94.33±0.47	87.78±0.27	86.36±0.57
RSAdapter[22]*	72.27±1.53	90.91±1.24	91.03±0.70	89.56±6.57	85.94±3.13	85.43±1.98
Ours	72.70±0.15	**92.31±0.16**	**92.22±0.40**	**94.77±0.74**	**88.00±0.29**	**86.55±0.18**

2) *EasyToHard [46]:* A multilevel framework that fuses global context and regional cues. A self-paced curriculum trains the model from easy to hard samples.
3) *SHRNet [21]:* A spatial hierarchical reasoning network using hash-based multiscale features. Question-guided relation modules model object interactions for fine-grained answering.
4) *UCAGAN [54]:* A context-aware attention network that alternates guidance across modalities. Cross-modal alignment is improved, and class imbalance is alleviated through a combined objective.
5) *SAMFPN [55]:* A scale-aware feature pyramid that aggregates global and local visual features. Co-attention and collaborative fusion support robust multiscale reasoning.
6) *Bi-Modal [56]:* A transformer-based method built on CLIP image and text encoders. Self-attention and cross-attention model dependencies and dual classifiers are fused for prediction.
7) *X-PFA [50]:* A frequency-domain transfer approach inserting primary frequency adapters into a pretrained backbone. It learns spectral structure with a small trainable parameter budget.
8) *RSAdapter [22]:* A parameter-efficient tuning framework attaching lightweight adapter blocks and linear transforms to multimodal transformers. Most backbone weights stay frozen, and adapters can be merged at inference.

1) Results on the RSVQA-LR Dataset: As shown in Table I, CRISP achieves the best OA of 86.55% and the best AA of 88.00% on RSVQA-LR. It improves over the classical RSVQA baseline by 6.47 points in OA and over EasyToHard by 3.46 points in OA. Compared with the strongest previous method, X-PFA, CRISP improves OA from 86.36% to 86.55% and AA from 87.78% to 88.00%. At the category level, the improvements are mainly reflected in Presence, Comparison, and Rural/Urban questions, while Count remains a relatively challenging category. In particular, compared with RSAdapter, CRISP improves Presence accuracy from 90.91% to 92.31% and Rural/Urban accuracy from 89.56% to 94.77%, indicating improved question-conditioned localization in large-scale Sentinel-2 scenes.

2) Results on the RSVQA-HR Test1 Dataset: Table II reports results on the in-distribution Test1 split. CRISP achieves the best OA of 85.82% and the best AA of 85.57%. Compared with the strongest competing overall result from SHRNet, CRISP improves OA from 85.39% to 85.82%

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TABLE II

COMPARISON WITH EXISTING METHODS ON THE RSVQA-HR TEST1 DATASET. ALL VALUES ARE REPORTED AS PERCENTAGES (%). * DENOTES OPEN-SOURCE CODE TESTED INDEPENDENTLY

Method	Count	Presence	Comparison	Area	AA	OA
RSVQA[45]*	68.24±0.15	90.86±0.03	89.35±0.10	85.53±0.07	83.49±0.08	83.67±0.02
EasyToHard[46]*	69.06±0.13	91.39±0.15	89.75±0.12	85.92±0.19	83.97±0.06	84.16±0.05
SHRNet[21]	70.04±0.15	92.45±0.11	91.68±0.09	86.35±0.13	85.13±0.08	85.39±0.05
UCAGAN[54]	**71.53±0.15**	91.10±0.09	88.56±0.35	86.83±0.30	84.51±0.07	84.56±0.18
SAMFPN[55]	69.19±0.14	91.99±0.15	91.44±0.09	85.84±0.19	84.61±0.14	84.94±0.13
Bi-Modal[56]	69.80±0.09	92.03±0.08	91.83±0.00	86.27±0.05	84.98±0.05	85.30±0.05
X-PFA[50]	70.12±0.08	92.05±0.07	91.85±0.07	86.32±0.04	85.08±0.03	85.36±0.05
RSAdapter[22]*	68.87±0.20	91.26±0.10	90.43±0.15	84.44±0.20	84.13±0.12	83.75±0.08
Ours	70.40±0.13	**92.52±0.17**	**92.06±0.04**	**87.29±0.08**	**85.57±0.07**	**85.82±0.07**

TABLE III

COMPARISON WITH EXISTING METHODS ON THE RSVQA-HR TEST2 DATASET. ALL VALUES ARE REPORTED AS PERCENTAGES (%). * DENOTES OPEN-SOURCE CODE TESTED INDEPENDENTLY

Method	Count	Presence	Comparison	Area	AA	OA
RSVQA[45]*	61.88±0.11	87.71±0.56	86.55±0.10	76.80±0.51	78.23±0.23	78.98±0.21
EasyToHard[46]*	61.95±0.08	87.97±0.06	87.68±0.23	78.62±0.23	79.06±0.15	79.29±0.15
SHRNet[21]	**63.42±0.14**	89.81±0.27	89.44±0.23	80.37±0.16	80.76±0.21	81.37±0.19
UCAGAN[54]	63.28±0.17	88.51±0.12	87.76±0.33	78.47±0.37	79.51±0.18	80.21±0.23
SAMFPN[55]	62.58±0.18	89.04±0.34	88.95±0.15	78.36±0.21	79.73±0.12	80.54±0.11
Bi-Modal[56]	63.06±0.11	89.37±0.21	89.62±0.29	80.12±0.39	80.54±0.16	81.23±0.15
X-PFA[50]	62.44±0.08	88.30±0.06	89.54±0.41	79.16±0.31	79.86±0.11	80.30±0.28
RSAdapter[22]*	62.86±0.12	88.78±0.44	88.87±0.87	80.48±0.49	80.08±0.35	80.61±0.40
Ours	62.94±0.06	**90.09±0.18**	**89.72±0.21**	**81.57±0.25**	**81.08±0.06**	**81.63±0.07**

and AA from 85.13% to 85.57%. At the category level, CRISP achieves the best results on Presence, Comparison, and Area, reaching 92.52%, 92.06%, and 87.29%, respectively. Compared with RSAdapter, it improves OA by 2.07 points and shows clear gains on Comparison and Area, indicating improved alignment between question intent and localized evidence in high-resolution aerial images.

3) Results on the RSVQA-HR Test2 Dataset: As shown in Table III, CRISP achieves the best OA of 81.63% and the best AA of 81.08% on the cross-region Test2 split. Compared with the strongest competing OA from SHRNet, CRISP improves OA from 81.37% to 81.63%, and also improves AA from 80.76% to 81.08%. Relative to RSAdapter, CRISP improves OA from 80.61% to 81.63% and yields notable gains on Comparison from 88.87% to 89.72%, while also improving Presence and Area. These results indicate that the proposed residual guidance and spatial realignment remain effective under cross-region evaluation, although the margin over the strongest baseline is moderate.

Overall, CRISP yields consistent improvements in OA and average per-type accuracy on RSVQA-LR and both RSVQA-HR test splits, with the largest gains observed on Presence, Comparison, Rural/Urban, and Area questions. The low standard deviations reported in Tables I–III further indicate that the performance of CRISP is stable under repeated runs. The Count category remains comparatively weaker, where CRISP is slightly inferior to the best competing approaches on all three splits. This behavior is consistent with the use of a frozen ViLT backbone operating on coarse image patches, without explicit object-centric modeling or density estimation. As a result, accurately enumerating multiple small and densely distributed objects remains more challenging than answering existence or relational questions. Even with this limitation, the improvements in OA and AA suggest that the proposed residual guidance and spatial realignment provide an effective parameter-efficient framework for RSVQA, particularly for presence, comparison, and region-related reasoning.

TABLE IV

ABLATION STUDY OF EACH VARIANT OF CRISP ON THE RSVQA-LR DATASETS. ALL VALUES ARE REPORTED AS PERCENTAGES (%)

Variant	Count	Presence	Comparison	Rural/Urban	AA	OA
Baseline	61.93	88.73	84.65	80.72	79.01	79.06
Baseline + CMRG	68.01	90.10	89.67	88.24	84.01	83.30
Baseline + ASR	67.59	90.08	89.30	88.00	83.75	83.13
CRISP(CMRG + ASR)	**72.70**	**92.31**	**92.22**	**94.77**	**88.00**	**86.55**

TABLE V

ABLATION STUDY OF EACH VARIANT OF CRISP ON THE RSVQA-HR TEST1 DATASETS. ALL VALUES ARE REPORTED AS PERCENTAGES (%)

Variant	Count	Presence	Comparison	Area	AA	OA
Baseline	68.34	91.40	89.60	84.95	83.57	83.83
Baseline + CMRG	69.32	92.03	91.35	86.39	84.90	85.17
Baseline + ASR	70.06	92.16	91.86	86.60	85.17	85.46
CRISP(CMRG + ASR)	**70.40**	**92.52**	**92.06**	**87.29**	**85.57**	**85.82**

TABLE VI

ABLATION STUDY OF EACH VARIANT OF CRISP ON THE RSVQA-HR TEST2 DATASETS. ALL VALUES ARE REPORTED AS PERCENTAGES (%)

Variant	Count	Presence	Comparison	Area	AA	OA
Baseline	62.03	87.96	86.75	76.45	78.30	79.10
Baseline + CMRG	62.28	88.05	89.17	79.33	79.70	80.42
Baseline + ASR	62.53	88.00	88.61	79.66	79.70	80.33
CRISP(CMRG + ASR)	**62.94**	**90.09**	**89.72**	**81.57**	**81.08**	**81.63**

D. Ablation Studies

To verify the effectiveness of the proposed CRISP framework and analyze the contribution of each component, we conducted extensive ablation studies on the RSVQA-LR and RSVQA-HR datasets.

1) Impact of Key Components: We first evaluate the individual contributions of the CMRG and ASR modules. We set the frozen ViLT backbone with a standard classifier as the Baseline. As reported in Tables IV–VI, combining both modules consistently outperforms the baseline and the single-module variants.

1) On the RSVQA-LR dataset (Table IV), the Baseline achieves an OA of 79.06%. Incorporating the CMRG module independently yields an OA of 83.30%, whereas the standalone ASR module achieves an OA of 83.13%. In contrast, the fully integrated CRISP framework attains the optimal performance, reaching the highest OA of 86.55% and AA of 88.00%.
2) On the RSVQA-HR datasets (Tables V and VI), a similar trend is observed. On RSVQA-HR Test1, the full model achieves the highest OA of 85.82% and AA of 85.57%. On the challenging Test2 split (Table VI), the full model achieves an OA of 81.63% and AA of 81.08%, surpassing the "Baseline + CMRG" and "Baseline + ASR" variants by 1.21% and 1.30% in OA, respectively.

These results indicate a complementary effect between the two modules. CMRG improves query-conditioned semantic guidance at the input stage, while ASR enhances localized spatial refinement within the Transformer layers.

2) Effectiveness of ASR: To validate the design of ASR, we analyze the contribution of the AGOG module. Using a frozen ViLT backbone with only the answer prediction head trained as the baseline setting, we evaluate three variants:

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TABLE VII

ABLATION STUDY OF EACH VARIANT OF ASR ON THE THREE RSVQA DATASETS. ALL VALUES ARE REPORTED AS PERCENTAGES (%)

Variant	RSVQA-LR		RSVQA-HR Test1		RSVQA-HR Test2	
	AA	OA	AA	OA	AA	OA
Baseline	79.01	79.06	83.57	83.83	78.30	79.10
Baseline + ASR (w/o AGOG)	83.22	82.79	84.67	85.03	79.64	80.30
Baseline + ASR	**83.75**	**83.13**	**85.17**	**85.46**	**79.70**	**80.33**

TABLE VIII

COMPARISON WITH REPRESENTATIVE OFFSET-BASED ADAPTATION MECHANISMS ON THE THREE RSVQA BENCHMARKS

Variant	RSVQA-LR		RSVQA-HR Test1		RSVQA-HR Test2	
	AA	OA	AA	OA	AA	OA
CMRG + Deformable Attention	86.20	84.72	85.06	85.34	80.32	80.43
CMRG + DCNv2	82.78	83.73	84.19	84.52	79.87	80.50
CMRG + ASR	**88.00**	**86.55**	**85.57**	**85.82**	**81.08**	**81.63**

1) Baseline; 2) Baseline + ASR (w/o AGOG), where the realignment branch is retained but the offsets are fixed rather than dynamically predicted via attention; and 3) Baseline + ASR, i.e., the full realignment module. As summarized in Table VII, ASR consistently improves performance over the Baseline, while removing AGOG leads to a consistent drop. For example, on RSVQA-LR, the OA increases from 79.06% for the Baseline to 83.13% with the full ASR, whereas removing AGOG reduces it to 82.79%. Similar trends are also observed on RSVQA-HR Test1 and Test2. These results suggest that the attention-guided mechanism contributes to more effective spatially variant offset prediction for localized feature realignment.

To further distinguish ASR from existing offset-based adaptation mechanisms, we compare it with two representative alternatives, namely Deformable Attention [57] and DCNv2 [58], under the same CMRG setting. As shown in Table VIII, CMRG + ASR achieves the best performance on all three benchmarks. Different from Deformable Attention, which changes the attention sampling process, ASR preserves the frozen Transformer attention structure and introduces a lightweight parallel residual branch for spatial feature realignment. Different from DCNv2, which performs convolutional offset-based sampling in local neighborhoods, ASR predicts attention-guided offsets for intermediate visual representations and conducts group-wise dynamic resampling. These results verify that ASR provides a more suitable spatial adaptation mechanism for frozen-backbone RSVQA.

3) Effectiveness of CMRG: We further investigate the design choices within the CMRG module by comparing our residual-guidance design against static guidance learning (fixed learnable guidance tokens), visual-only guidance learning, and question-only guidance learning. Table IX reports the results. On all three benchmarks, CMRG outperforms static, visual-only, and question-only guidance. For example, on RSVQA-LR, CMRG achieves an OA of 83.30%, compared with 82.62% for static guidance learning, 83.17% for visual-only guidance, and 83.13% for question-only guidance. On RSVQA-HR Test2, CMRG obtains 79.70% AA and 80.42% OA, clearly outperforming question-only guidance, which achieves 77.82% AA and 78.55% OA. This indicates that conditioning the guidance tokens on both the global visual summary and the question semantics helps the model better focus on image regions relevant to the query intent, and confirms that visual-textual fusion is more effective than using either modality alone for guidance generation.

TABLE IX

ABLATION STUDY OF EACH VARIANT OF CMRG ON THE THREE RSVQA DATASETS. ALL VALUES ARE REPORTED AS PERCENTAGES (%)

Variant	RSVQA-LR		RSVQA-HR Test1		RSVQA-HR Test2	
	AA	OA	AA	OA	AA	OA
Static Guidance Tokens	83.00	82.62	83.72	84.10	78.04	78.71
Visual-Only Guidance	83.76	83.17	84.63	84.94	79.68	80.36
Question-Only Guidance	83.83	83.13	84.64	84.95	77.82	78.55
CMRG	**84.01**	**83.30**	**84.90**	**85.17**	**79.70**	**80.42**

TABLE X

ABLATION STUDY ON DIFFERENT IMAGE/QUESTION SUMMARIZATION STRATEGIES FOR CMRG

Strategy	RSVQA-LR		RSVQA-HR Test1		RSVQA-HR Test2	
	AA	OA	AA	OA	AA	OA
Max	82.68	82.24	84.68	84.96	78.29	79.03
Mean	83.26	82.66	84.14	84.49	78.12	78.84
Ours	84.01	83.30	84.90	85.17	79.70	80.42

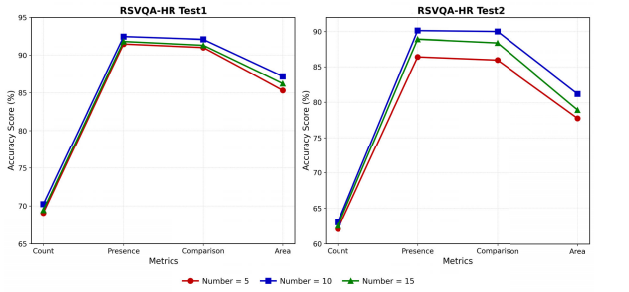


Fig. 3. Comparison of the accuracy of the complete CRISP model across different guidance lengths for various question types on two HR test datasets.

4) Impact of Summarization Strategy in CMRG: Compact pooled summaries are motivated by prior attention-based aggregation studies [59], [60], [61], [62] and by the sparse, background-dominated nature of RSVQA scenes. To further analyze why the pooled image and question summaries are used in CMRG, we compare different summarization strategies, including max pooling, mean pooling, and the adopted pooling strategy used in our method. Except for the summarization operator, all other components and training settings are kept unchanged. As reported in Table X, the adopted pooling strategy consistently outperforms max pooling and mean pooling on all three benchmarks. Specifically, it achieves 84.01% AA and 83.30% OA on RSVQA-LR, 84.90% AA and 85.17% OA on RSVQA-HR Test1, and shows a more pronounced advantage on the cross-region RSVQA-HR Test2 split, improving OA by 1.39% over max pooling and 1.58% over mean pooling. These results indicate that mean pooling may dilute sparse question-relevant evidence due to the large amount of background information in RSIs, while max pooling may be sensitive to isolated high-response tokens and ignore distributed semantic cues. In contrast, the adopted pooling strategy assigns different importance to visual and textual tokens during summarization, enabling CMRG to obtain more

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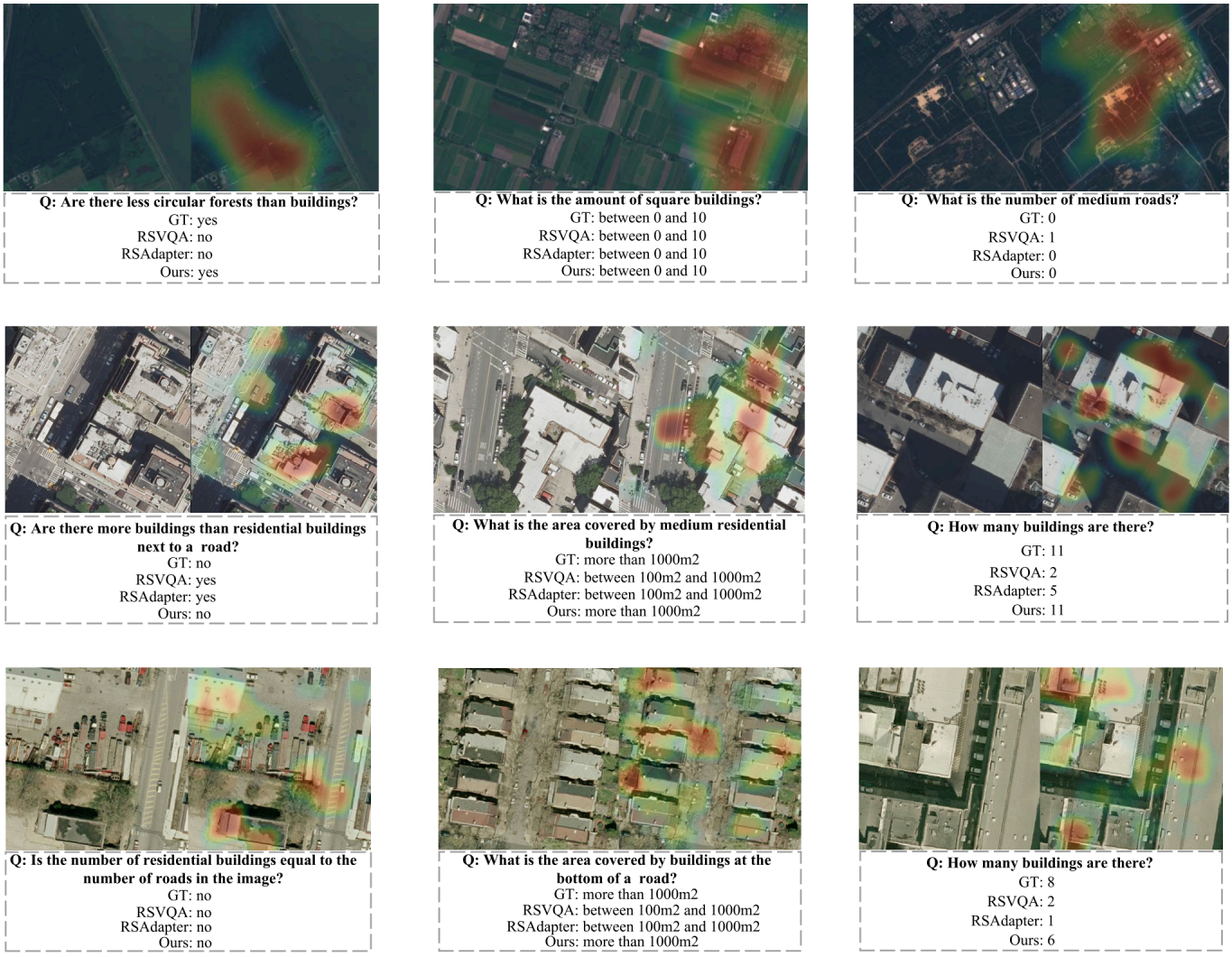


Fig. 4. Qualitative results with attention-map visualization on three RSVQA benchmarks. Each example presents the RSI, the question, the ground-truth answer, and the predictions from RSVQA, RSAdapter, and the proposed method. (Top), (Middle), and (Bottom) Rows correspond to RSVQA-LR, RSVQA-HR Test1, and RSVQA-HR Test2, respectively. The shown cases illustrate success and failure patterns across different spatial resolutions and scene layouts.

informative instance-level image and question summaries for guidance generation.

5) Sensitivity to Guidance Length: Finally, we analyze the impact of the number of guidance tokens (K) in CMRG. Fig. 3 illustrates the accuracy curves for $K = \{5, 10, 15\}$ on the RSVQA-HR Test1 and Test2 datasets. We observe that the model performance is relatively stable but generally peaks when the guidance length is set to 10. Too few tokens may not carry sufficient context-aware guidance, while too many may introduce redundancy. Consequently, we adopt $K = 10$ as the default setting.

E. Qualitative Analysis With Visualization

To provide intuitive insights into how CRISP handles the heterogeneity of RS scenes, we present qualitative comparisons in Fig. 4. We selected representative samples from the RSVQA-LR and RSVQA-HR datasets to evaluate the model's performance across three distinct reasoning capabilities: logical inference, spatial-geometric estimation, and dense object counting.

To further improve interpretability, we additionally overlay attention maps on the qualitative examples in Fig. 4. Warmer regions indicate higher attention responses associated with the question-answering process. The attention maps show that CRISP generally focuses on regions that are semantically relevant to the given question. For example, the model attends to forest and building regions for comparison questions, road regions for road-counting questions, and building clusters for area estimation and dense counting questions. These visualizations provide qualitative evidence that the proposed residual guidance helps promote question-relevant semantic focusing in heterogeneous RS scenes.

On RSVQA-LR, CRISP correctly answers questions requiring both object counting and semantic reasoning. For example, in scenes involving the number of medium roads or distinguishing rural from urban areas, our model produces the ground-truth answers while either RSVQA or RSAdapter fails. In the more complex question "Are there less circular forests than buildings?," CRISP successfully captures the relative quantities of irregular forest patches and building regions, whereas both baselines misclassify the relation. These cases

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TABLE XI
COMPARISON OF PARAMETER COUNTS AND COMPUTATIONAL EFFICIENCY ON THE RSVQA-LR DATASET. TOTAL PARAMETERS, TRAINABLE PARAMETERS, TRAINABLE RATIO, TRAINING TIME, AND TESTING TIME ARE REPORTED. "/" INDICATES THAT THE CORRESPONDING RESULT WAS NOT REPORTED AND COULD NOT BE REPRODUCED

Method	Parameters(M)	Trainable Parameters(M)	Trainable Ratio(%)	Train Time(s)	Test Time(s)
RSVQA	85.7	5.7	6.65	200	36
SHRNet	105.56	105.56	100	746	112
X-PFA	167.2	17.3	10.35	468	/
CRISP	131	20.4	15.57	195	51

highlight the benefit of instance-specific residual guidance that steers attention toward question-relevant structures (roads, settlements, and forest clusters), as well as the ability of ASR to refine heterogeneous spatial evidence.

The middle row illustrates high-resolution urban scenes from RSVQA-HR Test1. CRISP is able to recognize subtle differences in built-up areas, such as correctly estimating whether the area covered by buildings exceeds 1000 m^2, or identifying the presence of a residential building on the left of a road, where both RSVQA and RSAdapter overlook the target object. For questions about the number of medium commercial buildings, our method also produces the correct count while the baselines systematically overestimate. These examples suggest that combining CMRG and ASR improves both fine-grained spatial localization and relational reasoning in cluttered urban environments.

Finally, the bottom row presents challenging cross-region samples from RSVQA-HR Test2. In several cases, such as determining whether the number of residential buildings equals the number of roads, all methods succeed, indicating that the underlying ViLT backbone already captures some global layout cues. However, for more difficult counting questions, e.g., "How many buildings are there?," CRISP still underestimates the exact number of small objects, although its prediction (e.g., six buildings versus the ground-truth eight) is closer to reality than the much lower counts produced by RSVQA and RSAdapter. This behavior is consistent with our quantitative analysis, where the Count category remains the most challenging due to dense small objects and the absence of explicit object-centric modeling in the frozen backbone.

Overall, the visual comparisons corroborate the conclusions drawn from the ablation studies: instance-specific residual guidance helps the model focus on semantically relevant regions, while the ASR module refines localized spatial details and heterogeneous structures. Their combination enables CRISP to answer presence, comparison, and area-related questions more reliably than existing RSVQA baselines.

F. Efficiency and Parameter Analysis

To provide a practical efficiency reference, Table XI reports the total parameters, trainable parameters, trainable ratio, and measured training and testing time on the RSVQA-LR dataset for several representative methods. The trainable ratio is calculated as the number of trainable parameters divided by the total number of parameters, which provides a clearer view of parameter usage under the frozen-backbone setting. CRISP contains 131 M total parameters, among which 20.4 M parameters are trainable, corresponding to a trainable ratio of 15.57%. The number of trainable parameters in CRISP is substantially lower than that of SHRNet, which requires full model tuning with 105.56 M trainable parameters, corresponding to an 80.7% reduction in trainable parameters. In comparison, the classical RSVQA baseline updates 5.7 M parameters with a trainable ratio of 6.65%, while X-PFA updates 17.3 M parameters with a trainable ratio of 10.35%.

In terms of computational cost on RSVQA-LR, CRISP achieves a training time of 195 s, which is about 3.8× faster than SHRNet at 746 s and about 2.4× faster than X-PFA at 468 s. Its training time is also comparable to the RSVQA baseline at 200 s, suggesting that freezing the backbone effectively limits backpropagation overhead. During testing, CRISP requires 51 s, which is higher than RSVQA at 36 s due to the additional guidance generation and spatial realignment operations, but remains lower than SHRNet at 112 s. Overall, these results indicate that CRISP provides a reasonable tradeoff between model capacity, trainable parameter ratio, and computational cost on RSVQA-LR.

V. CONCLUSION

In this article, we proposed CRISP, a task-specific parameter-efficient adaptation framework for RSVQA that integrates the CMRG module and the ASR module on top of a frozen backbone to address the scale variation and sparse semantic distribution of RS imagery. By introducing query-conditioned semantic guidance at the input stage and performing ASR within intermediate layers, the proposed framework improves cross-modal alignment without the computational burden of full fine-tuning. Experimental results on RSVQA-LR and RSVQA-HR show that CRISP achieves strong overall performance and improves OA and AA over prior methods, especially on presence, comparison, and area/rural-urban question types. The gains on the HR benchmarks are moderate, and dense small-object counting remains challenging under the current frozen patch-based backbone. These findings indicate that residual guidance and spatial realignment are effective for semantic alignment and region-related reasoning in RSVQA, while counting-oriented extensions are still needed for fine-grained enumeration tasks.

In future work, the limitations on Count questions will be addressed by incorporating lightweight counting-aware cues to better handle small and densely distributed targets. We will also investigate these challenges from a mathematically grounded perspective [63], [64]. In addition, the framework will be extended to broader sensors and regions, and more efficient implementations of the spatial realignment branch will be explored to further improve generalization and deployment efficiency.

REFERENCES

[1] J. Bu et al., "Machine learning methods for Earth observation and remote sensing using spaceborne GNSS reflectometry: Current status, challenges, and future prospects," *IEEE Geosci. Remote Sens. Mag., Replaces Newslett.*, vol. 13, no. 3, pp. 8–71, Sep. 2025.

[2] T. Gao, Z. Li, Y. Wen, T. Chen, Q. Niu, and Z. Liu, "Attention-free global multiscale fusion network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5603214.

[3] D. Liu, J. Zhang, Y. Qi, Y. Wu, and Y. Zhang, "Tiny object detection in remote sensing images based on object reconstruction and multiple receptive field adaptive feature enhancement," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5616213.

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[4] X. Yuan, G. Cheng, R. Yao, and J. Han, "Semantic differentiation aids oriented small object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 6, pp. 5966–5979, Jun. 2025.

[5] Z. Huang et al., "T2EA: Target-aware Taylor expansion approximation network for infrared and visible image fusion," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 5, pp. 4831–4845, May 2025.

[6] X. Wu, T.-Z. Huang, L.-J. Deng, and T.-J. Zhang, "Dynamic cross feature fusion for remote sensing pansharpening," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 14687–14696.

[7] J. Ning, Z. Ren, B. Hou, W. Li, and L. Jiao, "Deep geospatio-semantic guided network with pseudo-label consistency for domain-adaptive remote sensing segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5634517.

[8] E. Zhu, Z. Chen, D. Wang, H. Shi, X. Liu, and L. Wang, "UNetMamba: An efficient UNet-like mamba for semantic segmentation of high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[9] X. Pu, H. Jia, L. Zheng, F. Wang, and F. Xu, "ClassWise-SAM-adapter: Parameter-efficient fine-tuning adapts segment anything to SAR domain for semantic segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 4791–4804, 2025.

[10] T. Luo, H. Li, M. Gong, Y. Zhou, and A. K. Qin, "STEAM: Style transfer enabled adversarial attack with attention mechanism on remote sensing image scene classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 12, pp. 12524–12537, Dec. 2025.

[11] W. Miao, J. Geng, and W. Jiang, "Multigranularity decoupling network with pseudolabel selection for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5603813.

[12] Z. Li et al., "SSFNet: Spectral–spatial fusion network for hyperspectral remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5508511.

[13] Y. Zhang, J. Zhen, T. Liu, Y. Yang, and Y. Cheng, "Adaptive differentiation Siamese fusion network for remote sensing change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[14] T. Zhu et al., "FTA-Net: Frequency-temporal-aware network for remote sensing change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 3448–3460, 2025.

[15] H. Zhang, K. Chen, C. Liu, H. Chen, Z. Zou, and Z. Shi, "CDMamba: Incorporating local clues into mamba for remote sensing image binary change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4405016.

[16] J. Li, D. Li, S. Savarese, and S. Hoi, "Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 19730–19742.

[17] C. Zhang, Z. Ren, B. Hou, J. Meng, W. Li, and L. Jiao, "Interactive concept network enhanced transformer for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4401416.

[18] Y. Zhao, M. Zhang, B. Yang, Z. Zhang, J. Kang, and J. Gong, "LuoJiaHOG: A hierarchy oriented geo-aware image caption dataset for remote sensing image–text retrieval," *ISPRS J. Photogramm. Remote Sens.*, vol. 222, pp. 130–151, Apr. 2025.

[19] Q. Lin, S. Wang, X. Ye, R. Wang, R. Yang, and L. Jiao, "CLIP-based grid features and masking for remote sensing image captioning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 2631–2642, 2025.

[20] S. Lobry and D. Tuia, "*Visual question answering on remote sensing images*," in *Advances in Machine Learning and Image Analysis for GeoAI*. Amsterdam, The Netherlands: Elsevier, 2024, pp. 237–254.

[21] Z. Zhang et al., "A spatial hierarchical reasoning network for remote sensing visual question answering," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4400815.

[22] Y. Wang and P. Ghamisi, "RSAdapter: Adapting multimodal models for remote sensing visual question answering," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5628313.

[23] Z. Hu et al., "Efficient grounding DINO: Efficient cross-modality fusion and efficient label assignment for visual grounding in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5609414.

[24] Y. Zhao, Y. Chen, R. Yao, S. Xiong, and X. Lu, "Context-driven and sparse decoding for remote sensing visual grounding," *Inf. Fusion*, vol. 123, Nov. 2025, Art. no. 103296.

[25] Y. Zhan, Z. Xiong, and Y. Yuan, "RSVG: Exploring data and models for visual grounding on remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5604513.

[26] Q. Ma, J. Pan, and C. Bai, "Direction-oriented visual–semantic embedding model for remote sensing image–text retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4704014.

[27] A.-A. Liu et al., "Text-guided knowledge transfer for remote sensing image-text retrieval," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[28] C. Zheng et al., "Cross-modal progressive perspective matching network for remote sensing image-text retrieval," *IEEE Trans. Multimedia*, vol. 27, pp. 3966–3978, 2025.

[29] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.

[30] J. Li, D. Li, C. Xiong, and S. Hoi, "BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 12888–12900.

[31] W. Kim, B. Son, and I. Kim, "ViLT: Vision-and-language transformer without convolution or region supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 5583–5594.

[32] X. Li, Y. Lu, J. Cao, Y. Ma, Z. Li, and Y. Zhou, "CATCH: A modular cross-domain adaptive template with hook," 2025, *arXiv:2510.26582*.

[33] H. Li et al., "RSI-CB: A large-scale remote sensing image classification benchmark using crowdsourced data," *Sensors*, vol. 20, no. 6, p. 1594, Mar. 2020.

[34] C. Chappuis, V. Zermatten, S. Lobry, B. Le Saux, and D. Tuia, "Prompt-RSVQA: Prompting visual context to a language model for remote sensing visual question answering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 1371–1380.

[35] R. Liu, C. Li, Y. Ge, T. H. Li, Y. Shan, and G. Li, "BT-adapter: Video conversation is feasible without video instruction tuning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 13658–13667.

[36] J.-M.-O. Steitz and S. Roth, "Adapters strike back," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 23449–23459.

[37] Z. Zhang et al., "Language-driven visual consensus for zero-shot semantic segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 4, pp. 3185–3195, Apr. 2025.

[38] Z. Lu, J. Bai, X. Li, Z. Xiao, and X. Wang, "Task-to-instance prompt learning for vision-language models at test time," *IEEE Trans. Image Process.*, vol. 34, pp. 1908–1920, 2025.

[39] Q. Sun et al., "GRASP: Guided region-aware sparse prompting for adapting MLLMs to remote sensing," 2026, *arXiv:2601.17089*.

[40] S. Antol et al., "VQA: Visual question answering," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 2425–2433.

[41] W. Chen, X. Huang, Z. Liu, J. Liu, and L. Yo, "RK-VQA: Rational knowledge-aware fusion-in-decoder for knowledge-based visual question answering," *Inf. Fusion*, vol. 118, Jun. 2025, Art. no. 102969.

[42] D. Peng and Z. Li, "Unbiased VQA via modal information interaction and question transformation," *Pattern Recognit.*, vol. 162, Jun. 2025, Art. no. 111394.

[43] J. Zhang and Y. Zhang, "DocRouter: Prompt guided vision transformer and mixture of experts connector for document understanding," *Inf. Fusion*, vol. 122, Oct. 2025, Art. no. 103206.

[44] J. Pfeiffer et al., "AdapterHub: A framework for adapting transformers," 2020, *arXiv:2007.07779*.

[45] S. Lobry, D. Marcos, J. Murray, and D. Tuia, "RSVQA: Visual question answering for remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8555–8566, Dec. 2020.

[46] Z. Yuan, L. Mou, Q. Wang, and X. X. Zhu, "From easy to hard: Learning language-guided curriculum for visual question answering on remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5623111.

[47] A. Sarkar, T. Chowdhury, R. R. Murphy, A. Gangopadhyay, and M. Rahnemoonfar, "SAM-VQA: Supervised attention-based visual question answering model for post-disaster damage assessment on remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4702716.

[48] J. Feng, S. Li, and S. Dong, "Hierarchical feature integration and fusion for remote sensing visual question answering," *Displays*, vol. 90, Dec. 2025, Art. no. 103099.

[49] L. Zhu, X. Su, J. Tang, Y. Hu, and Y. Wang, "View-based knowledge-augmented multimodal semantic understanding for optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5608333.

[50] Z. Zhang et al., "Frequency domain transfer learning for remote sensing visual question answering," *Expert Syst. Appl.*, vol. 291, Oct. 2025, Art. no. 128395.

[51] K. Kuckreja, M. S. Danish, M. Naseer, A. Das, S. Khan, and F. S. Khan, "GeoChat: Grounded large vision-language model for remote sensing," in *Proc. IEEE/CVF Conf. Comput. Vis. pattern Recognit.*, Nov. 2024, pp. 27831–27840.

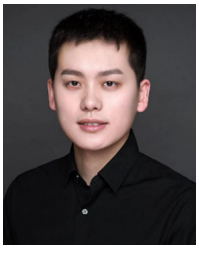
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[52] C. Pang et al., "VHM: Versatile and honest vision language model for remote sensing image analysis," in *Proc. AAAI Conf. Artif. Intell.*, vol. 39, 2025, pp. 6381–6388.
[53] J. Luo et al., "SkySenseGPT: A fine-grained instruction tuning dataset and model for remote sensing vision-language understanding," 2024, *arXiv:2406.10100*.
[54] J. Feng, E. Tang, M. Zeng, Z. Gu, P. Kou, and W. Zheng, "Improving visual question answering for remote sensing via alternate-guided attention and combined loss," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 122, Aug. 2023, Art. no. 103427.
[55] J. Feng and H. Wang, "A multi-scale contextual attention network for remote sensing visual question answering," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 126, Feb. 2024, Art. no. 103641.
[56] Y. Bazi, M. M. A. Rahhal, M. L. Mekhalfi, M. A. A. Zuair, and F. Melgani, "Bi-modal transformer-based approach for visual question answering in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4708011.
[57] Z. Xia, X. Pan, S. Song, L. E. Li, and G. Huang, "Vision transformer with deformable attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4794–4803.
[58] R. Wang et al., "DCN V2: Improved deep & cross network and practical lessons for web-scale learning to rank systems," in *Proc. Web Conf.*, Apr. 2021, pp. 1785–1797.
[59] M. V. Koroteev, "BERT: A review of applications in natural language processing and understanding," 2021, *arXiv:2103.11943*.
[60] J. Lee, Y. Lee, J. Kim, A. R. Kosiorek, S. Choi, and Y. W. Teho, "Set Transformer: A framework for attention-based permutation-invariant neural networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2019, pp. 3744–3753.
[61] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 30, 2017, pp. 5998–6008.
[62] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[63] Z. Huang, Y. Yang, J. Liu, Q. Li, Y. Shi, and Y. Zhang, "RUAN: Residual U-shaped attention network to approximate Taylor expansion for image denoising," *Digit. Signal Process.*, vol. 168, Jan. 2026, Art. no. 105505.
[64] Z. Huang, W. Hu, Z. Zhu, Q. Li, and H. Fang, "TMSF: Taylor expansion approximation network with multi-stage feature representation for optical flow estimation," *Digit. Signal Process.*, vol. 162, Jul. 2025, Art. no. 105157.



Changhui Xu received the B.S. degree in software engineering from Henan University of Science and Technology, Luoyang, China, in 2023. He is currently pursuing the Ph.D. degree in computer science and technology with Xidian University, Xi'an, China.

He is currently a member of the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, Xi'an University. His research interests encompass visual language models and visual question-answering systems.



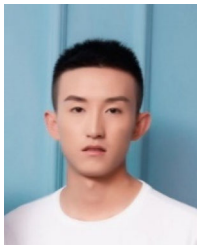
Zhongle Ren (Member, IEEE) received the B.S. degree in electronic and information engineering and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, China, in 2013 and 2020, respectively.

He is currently a Lecturer with the School of Artificial Intelligence, Xidian University. His research interests include large-scale remote sensing pretraining models and synthetic aperture radar (SAR) images' understanding and interpretation.



Biao Hou (Senior Member, IEEE) received the B.S. and M.S. degrees in mathematics from Northwest University, Xi'an, China, in 1996 and 1999, respectively, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, in 2003.

Since 2003, he has been with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China, Xidian University, where he is currently a Professor. His research interests include compressive sensing and synthetic aperture radar image interpretation.



Cheng Zhang received the M.S. degree in computer science and technology from Xi'an University of Technology, Xi'an, China, in 2023. He is currently pursuing the Ph.D. degree with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China, Xidian University, Xi'an.

His research interests include remote sensing image interpretation, vision-and-language reasoning, and multimodal learning.



Jiawei Ning received the B.S. degree in computer science and technology from the Mingde College, Northwestern Polytechnical University, Xi'an, China, in 2014, and the master's degree in computer science and engineering from the School of Computer Science and Engineering, Xi'an University of Technology, Xi'an, in 2022. He is currently pursuing the Ph.D. degree with the School of Artificial Intelligence, Xidian University, Xi'an.

His research interests include transfer learning and remote sensing image semantic segmentation.



Weibin Li (Senior Member, IEEE) received the B.S. and M.S. degrees in mathematics from Northwest University, Xi'an, China, in 1998 and 2000, respectively, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, in 2004.

He is currently the Director of the Artificial Intelligence Laboratory, Hangzhou Institute of Technology, Xidian University, where he is currently a Professor. His research interests include remote sensing image processing and high-precision positioning technology and applications.



Licheng Jiao (Life Fellow, IEEE) received the B.S. degree from Shanghai Jiao Tong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

Since 1992, he has been a Professor with the School of Electronic Engineering, Xidian University, Xi'an, where he is currently the Director of the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China, and the International Research Center of Intelligent Perception and Computation. His research interests include intelligent information processing, image processing, machine learning, and pattern recognition.

Dr. Jiao is a member of the IEEE Xi'an Section Execution Committee, the President of the Computational Intelligence Chapter, the IEEE Xi'an Section, and the IET Xi'an Network, the Chairperson of the Awards and Recognition Committee, the Vice Board Chairperson of Chinese Association of Artificial Intelligence, a Councilor of Chinese Institute of Electronics, a Committee Member of Chinese Committee of Neural Networks, and an Expert of the Academic Degrees Committee of the State Council.

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